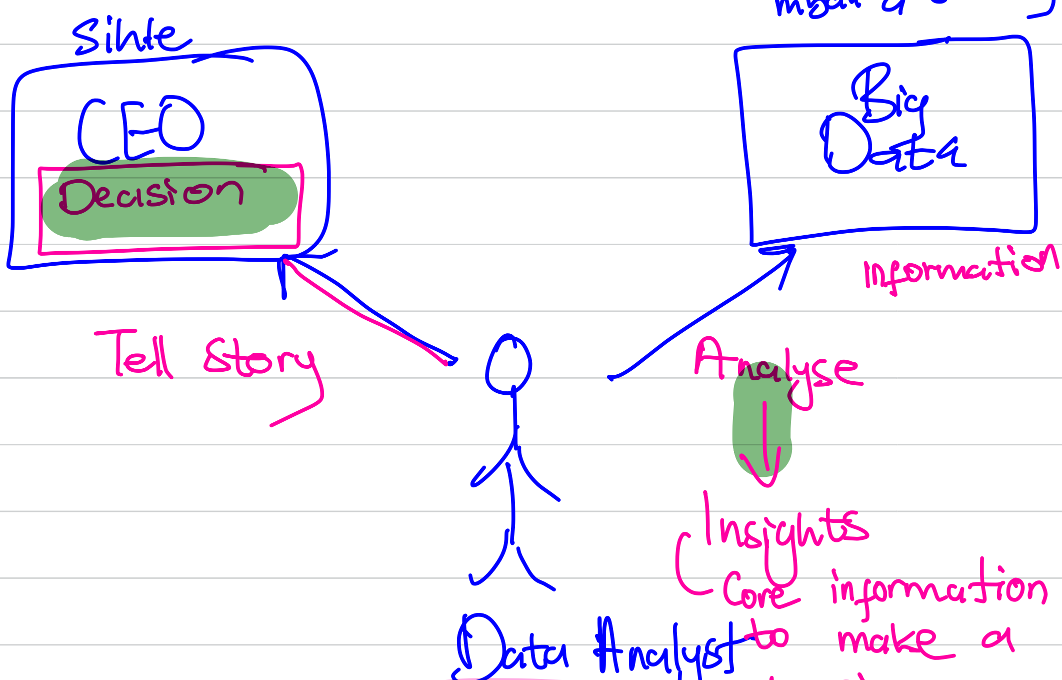
Data

└ is information

- You need to be able to tell a story from your data
- you need to drive/Extract actionable insights from your data
  └ Drive Business Decisions



Relational data base ⌒ structured data that can be related/joined together to become.

In a Relational database we store structured data that (dimension tables & fact tables) that can be connected to one another using primary key and foreign key

Primary key is a unique identifier that differentiate each component from the table e.g ID number

customers

Cust_id	name	Age	City
1	Rotimi	28	NY
2	Jeff	22	LA
3	Mark	18	MG

transactions

order_id	Cust_id	Product_id	Rev
324	2	104	70
333	1	102	80
401	3	101	100

- fact table
- has foreign key
- contains numbers
- used for analysis

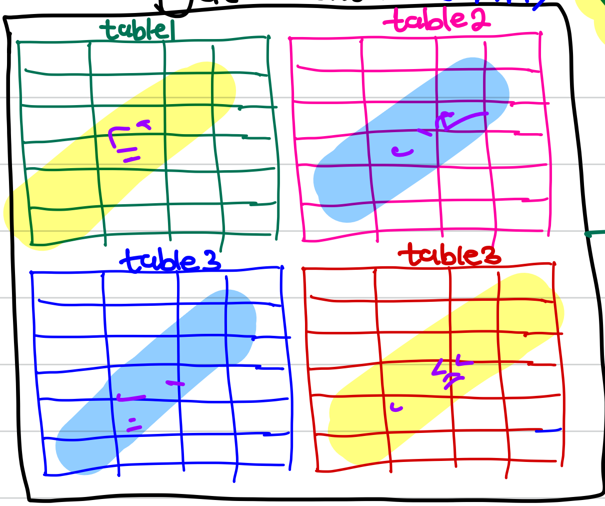
Products Dim-table

product_id	product_category	Product_name
101	fruits	Banana
102	fruits	Apple
103	Electronics	laptop
104	Clothing	uzzi jeans

Foreign key - is a primary key from one table that is found in another table

Dim_table - Dimension table contains descriptive information
- A Dim table does not consistently change

Datawarehouse (OLAP)



we use a DMS to access the data

connected to our data

Data Management system (DMS)

in a DMS we code

Why do we need to code?

- Data Inspection / Exploration
  - to see what data is available
- Data Exploratory Analysis
  - To understand the relationship between our data points
- feature Building
  - creating feature used in ML for predictions

we code to retrieve, to amend, to update, to manipulate, to clean, to aggregate, to join, to delete, to update with the goal of Extracting actionable insights

- Summarized table
- Aggregated table
- Combined table
- Unified table
- Insight table